

*MA.7.DP.1.2***Benchmark**

MA.7.DP.1.2 **Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results and draw conclusions about the two populations.**

Benchmark Clarifications:

Clarification 1: Graphical representations are limited to histograms, line plots, box plots and stem-and-leaf plots.

Clarification 2: The measure of center is limited to mean and median. The measure of variation is limited to range and interquartile range.

Connecting Benchmarks/Horizontal Alignment

- MA.6.DP.1.2
- MA.7.DP.1.1
- MA.7.NSO.2.2, MA.7.NSO.2.3

Terms from the K-12 Glossary

- Box Plot
- Data
- Histogram
- Interquartile Range (IQR)
- Line Plot
- Mean
- Measures of Center
- Measures of Variability
- Median
- Range (of data set)
- Stem-and-Leaf Plot

Vertical Alignment**Previous Benchmarks**

- MA.5.DP.1.2

Next Benchmarks

- MA.8.DP.1.1

Purpose and Instructional Strategies

In previous courses, students interpreted numerical data, with whole-number values, represented with tables or line plots by determining the mean, median, mode and range. In grade 6 accelerated, students will calculate and interpret mean, median, mode and range. They use those calculations to make comparisons, interpret results and draw conclusions about two populations. In future courses, students will learn how to interpret the main features of line graphs and lines of fit.

- Instruction includes cases where students need to calculate measures of center and variation in order to interpret them.
- Instruction includes having students collect their own data for analysis. Students make connections to data they are interested in. (*MTR.1.1, MTR.7.1*).
- Students should not be expected to classify differences between data sets as “significant” or “not significant.”

- Data representations can be shown as a two-sided stem-and-leaf plot and multiple box plots on the same scale.
- Data representations should include titles, labels and a key as appropriate.

Common Misconceptions or Errors

- Some students may incorrectly believe a histogram with greater variability in the heights of the bars indicates greater variability of the data set.
- Students may not recognize when to use a stem-and-leaf plot or may not be able to read a two-sided stem-and-leaf plot.
- Students may not be able to explain their choice of the most appropriate measures of center and variability based on the given data.
- Students may think that the presence of one or more outliers leads to an automatic choice (median, IQR) for the measures of center and variation.

Strategies to Support Tiered Instruction

- Instruction includes explaining the difference between variability in the heights of the bars of histograms, and the actual variability of the data set.
- Teacher provides instruction on how to use different types of data displays to show two sets of data at the same time. Teachers co-create an anchor chart explaining the different parts of each display with explanations on when and how to use each of them.
 - For example, the teacher can provide students with a two-sided stem-and-leaf plot with the “stem” in the middle and “leaves” on either side, each displaying the two data sets.

Data Set A		Data Set B
1, 2, 3	0	
4, 5, 5, 8	1	
5, 8, 9	2	2, 4, 8
	3	0, 0, 0, 6, 9
	4	3, 4, 9

Key: 1|0 = 1.0

- For example, the teacher can provide students with two line plots or two box plots on the same number line. Plots can be given in different colors to show the different data sets.
- Teacher provides instruction on which measure of center and variation should be used, making sure to include what to do when an outlier is present.
- Teacher facilitates discussion on the different measures of center and variability and how to know when to use each one. Use a graphic organizer to compare the different measures of center and variability to assist students in deciding when to use them.
- Instruction includes co-creating an anchor chart with different data displays containing visual representations and explanations of when and how to use them.

Instructional Tasks

Instructional Task 1 (MTR.1.1, MTR.7.1)

A group of students in the book club are debating whether high school juniors or seniors spend more time on homework. A random sampling of juniors and seniors at the local high school were surveyed about the average amount of time they spent per night on homework. The results are listed in the table below.

Average Amount of Time on Homework Per Night (in minutes)

Juniors	125	90	95	80	100	100	120	85	95	120	95	85
Seniors	135	70	65	100	35	175	80	80	90	65	120	60

Part A. Calculate and compare the measures of center for the data sets.

Part B. Calculate and compare the variability in each distribution.

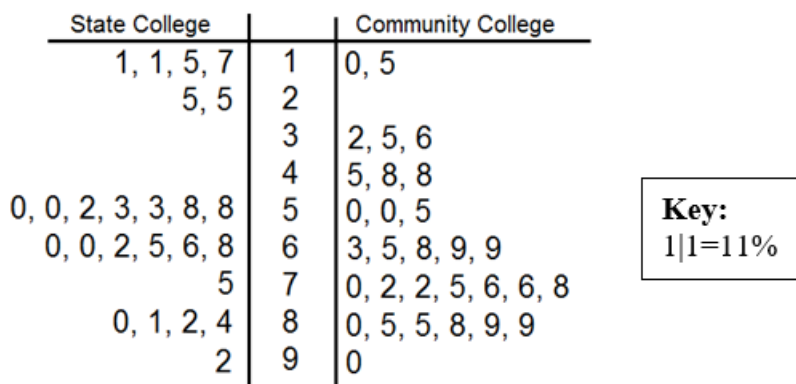
Part C. Does the data support juniors or seniors spending more time on homework?

Explain your reasoning.

Instructional Items

Instructional Item 1

High schools around the state of Florida were asked what percentage of students in their graduating class would be attending a state college and what percentage would be attending a community college. The results are provided in the graph below.



Is a student more likely to go to a state or community college? Which choice has more variability?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*